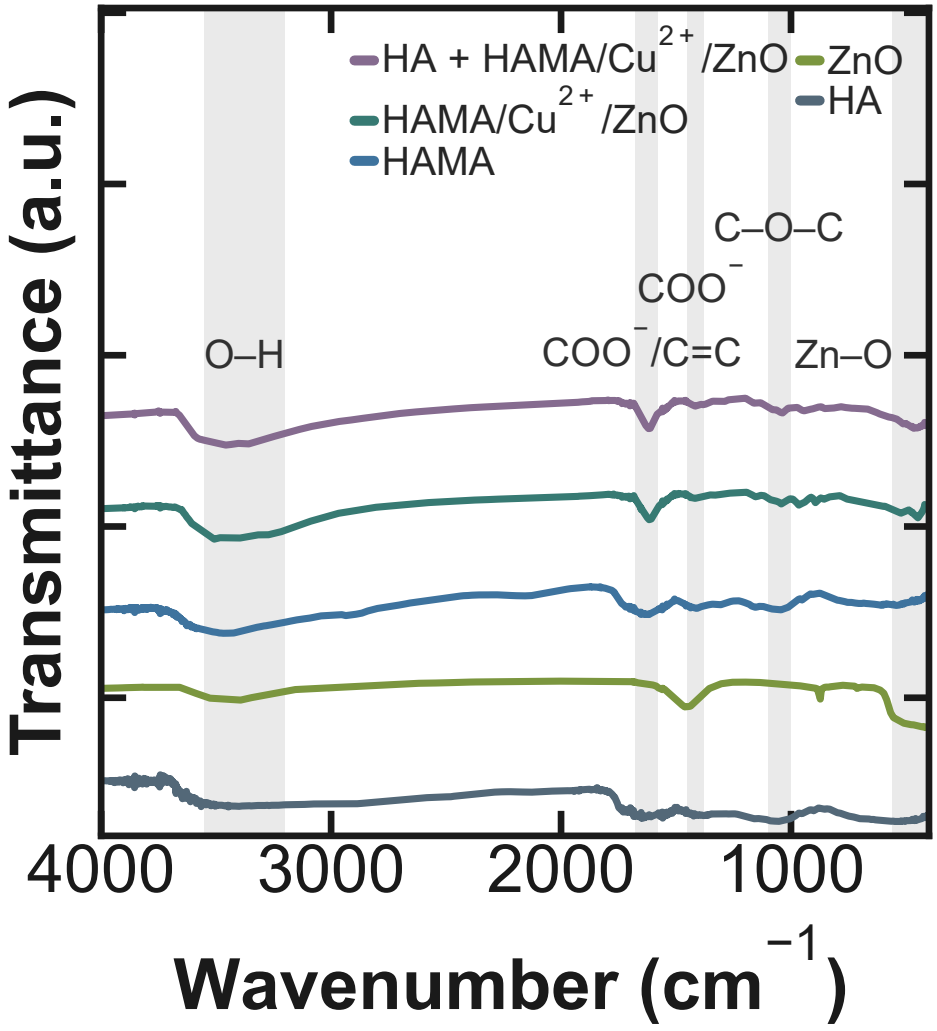
HA + HAMA/Cu2+ /ZnO
HAMA/Cu2+ /ZnO
HAMA
ZnO
HA
C–O–C
COO−
COO−/C=C
O–H
Zn–O
Transmittance (a.u.)
4000
3000
2000
1000
Wavenumber (cm−1)